Supervised learning

Artificial Intelligence
CSC 480
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Adapted with permission from
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Contents

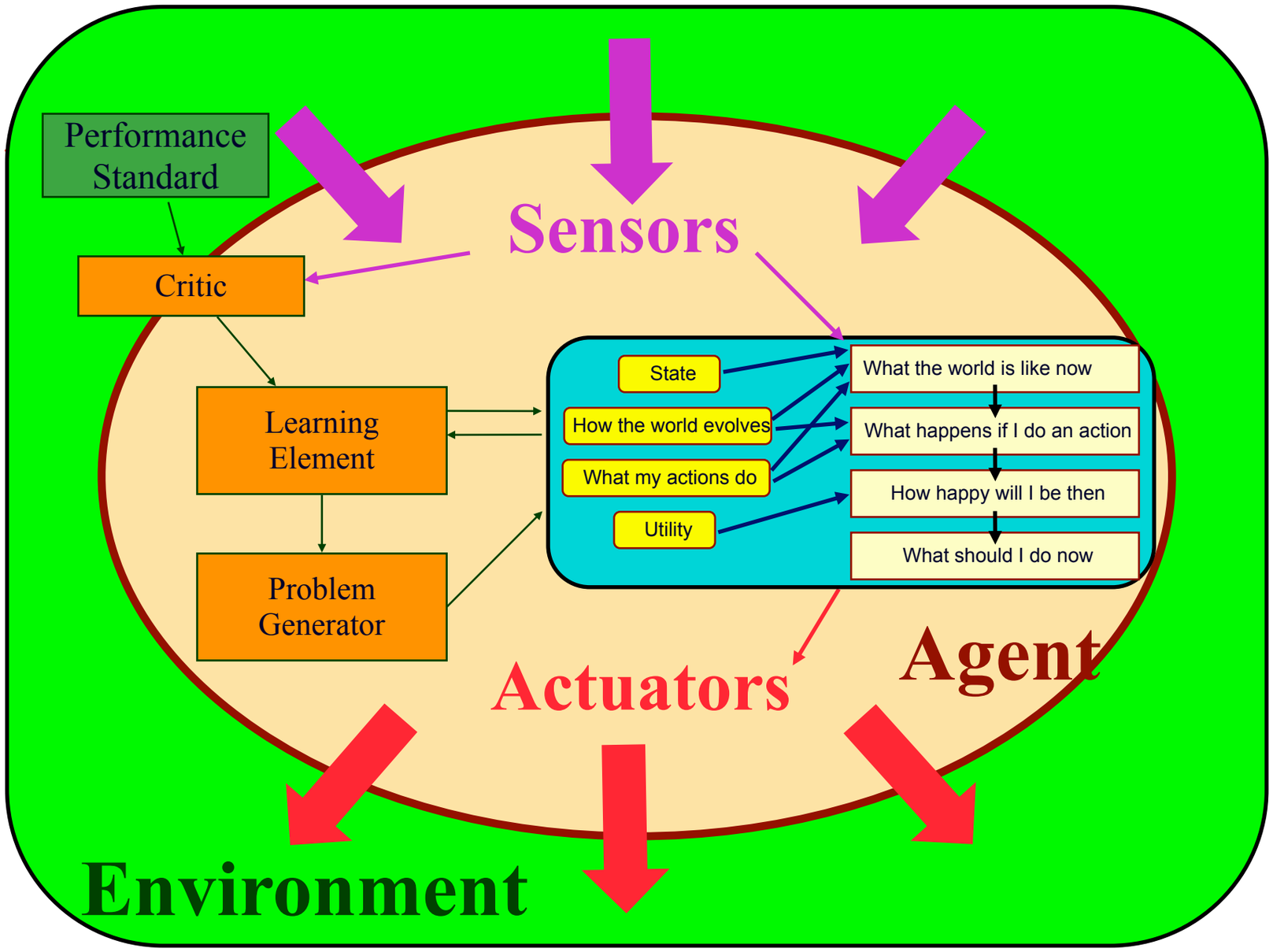
- Learning in intelligent agents
- Supervised learning and other forms of learning
- Eager and lazy learning
- k-Nearest Neighbor
- Decision Trees
- Perceptrons

Why learning?

Three main reasons:

- Designers might not anticipate all situations the agent might face (e.g. different mazes)
- Designers might not anticipate all changes over time to the environment (e.g. game patches, law changes)
- Designers might have no idea how to program a solution (e.g. face recognition)

A learning agent (AIMA ch.2)



What could be learned?

```
Welcome to
                EEEEEE  LL      IIII   ZZZZZZ   AAAAA
                EE      LL       II       ZZ   AA   AA
                EEEEE   LL       II     ZZZ    AAAAAAA
                EE      LL       II    ZZ      AA   AA
                EEEEEE  LLLLLL  IIII ZZZZZZ    AA   AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.


ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

What could be learned?

- What words most likely follow other words
- How syntax works in a language
- How to detect emotion in a sentence
- How to map words from one language to another
- Speech recognition / synthesis
- How to look up basic facts online (bitcoin price)

What could be learned?



What could be learned?

- How to drive from point A to point B, without hitting pedestrians or breaking laws
- What a human (or a cat, or bush) looks like
- The meaning of each traffic sign
- Which routes from A to B are actually fastest (beyond mere distance calculation)
- How far back you want your seat, temperature for the AC, favorite radio channel…
- Estimating distances

What could be learned?

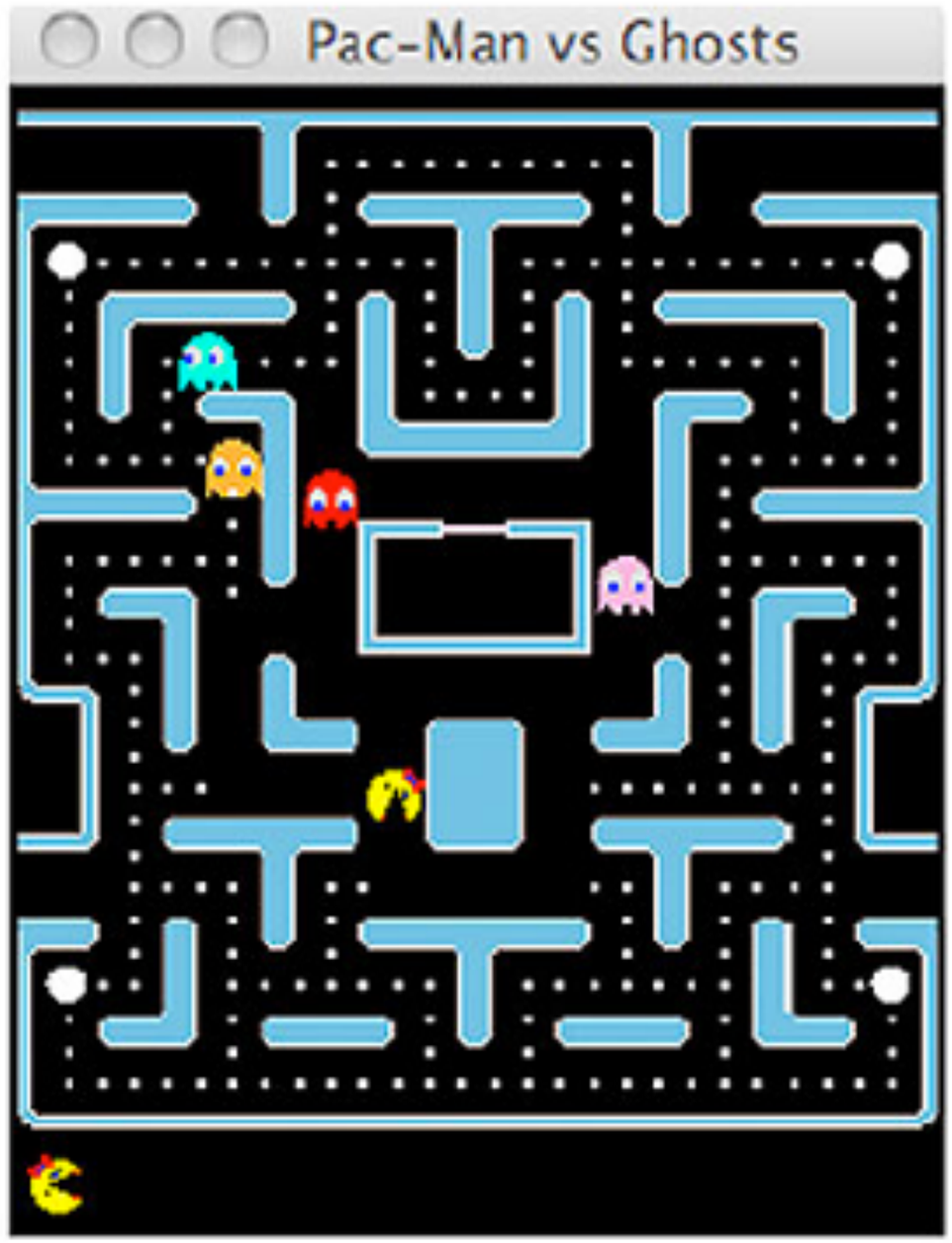


What could be learned?

- How much a board position is worth
- What action to take in a specific situation
- What action a particular person would take in a specific situation
- Who is likely to win a game between two people, and how long it takes

What could be learned?



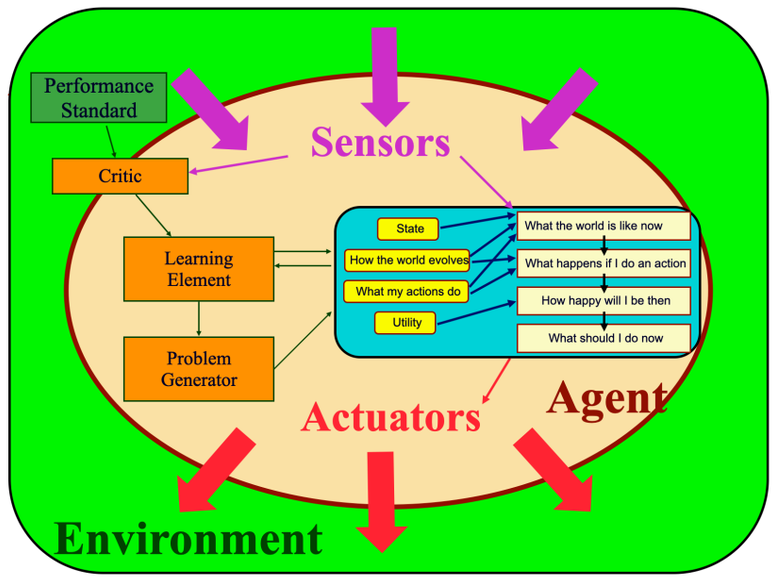


What could be learned?

- How ghosts move
- The value of a state
- Dangerous positions
- What action to take in a specific situation

What could be learned?

In general, <u>anything in the blue box</u>



- Mapping from states to actions
- Relevant properties of a partially observable world
- How the world evolves
- The desirability (utility) of each state
- The desirability of each action
- Goals to be pursued

Types of learning

- **Supervised learning**
  Learning to predict or classify labels based on labeled input data

- **Unsupervised learning**
  Finding patterns in unlabeled data

- **Reinforcement learning**
  Learning well-performing behavior from state observations and rewards

Meme Time!

- **"Pure" Reinforcement Learning** (cherry)
  - The machine predicts a scalar reward given once in a while.
  - **A few bits for some samples**
- **Supervised Learning** (icing)
  - The machine predicts a category or a few numbers for each input
  - Predicting human-supplied data
  - **10→10,000 bits per sample**
- **Unsupervised/Predictive Learning** (cake)
  - The machine predicts any part of its input for any observed part.
  - Predicts future frames in videos
  - **Millions of bits per sample**
- **(Yes, I know, this picture is slightly offensive to RL folks. But I'll make it up)**



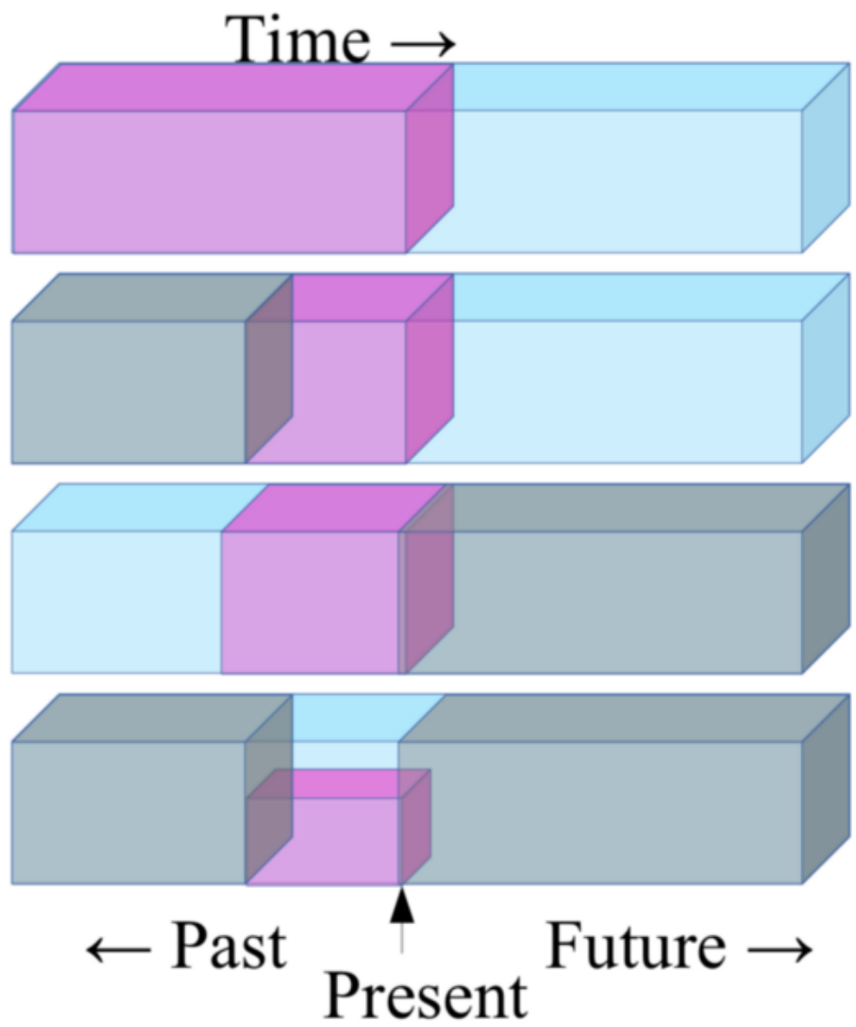
Source: Yann LeCun. Retrieved from https://medium.com/syncedreview/yann-lecun-cake-analogy-2-0-a361da560dae (access Nov 15 2021)

Self-Supervised Learning

Self-Supervised Learning

Y. LeCun

- **Predict any part of the input from any other part.**
- **Predict the future from the past.**
- **Predict the future from the recent past.**
- **Predict the past from the present.**
- **Predict the top from the bottom.**
- **Predict the occluded from the visible**
- **Pretend there is a part of the input you don't know and predict that.**



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